

Wildlife Detection Model Training & Deployment Manual (Roboflow + Google Colab + YOLOv8)

Complete User Guide

This manual explains dataset preparation in Roboflow, class modification, dataset export, Google Drive setup, YOLOv8 training in Google Colab, model export, and deployment preparation for RDK X5.

Step 1: Create / Open Project in Roboflow

1. Login to Roboflow.
2. Create a new Object Detection project.
3. Upload all wildlife images.
4. Verify annotations for each image.
5. Save the project.

Step 2: Merge Datasets

1. Upload images from multiple sources into the same project.
2. Review duplicate classes.
3. Ensure annotations are correctly mapped.
4. Generate a new project version after merging.

Step 3: Modify Classes

1. Open Classes section.
2. Rename classes if required.
3. Merge duplicate classes.
4. Delete unwanted classes.
5. Verify final class list before generating dataset version.

Step 4: Generate Dataset Version

Recommended:

- Preprocessing: Auto-Orient
- Resize: 640 x 640
- Augmentation: Flip, Rotate, Brightness (optional)

Generate a new version after selecting settings.

Step 5: Export Dataset

1. Open generated dataset version.
2. Click Export.
3. Select YOLOv8 format.
4. Download dataset ZIP file.

Step 6: Upload Dataset to Google Drive

1. Open Google Drive.
2. Create folder 'dataset'.
3. Upload exported ZIP file.

Step 7: Open Google Colab

1. Go to Google Colab.
2. Runtime → Change Runtime Type.
3. Select GPU (T4 preferred).

Step 8: Install Ultralytics

Code:

```
!pip install ultralytics
```

Step 9: Mount Google Drive

Code:

```
from google.colab import drive
drive.mount('/content/drive')
```

Step 10: Unzip Dataset

Code:

```
!unzip "/content/drive/MyDrive/dataset/dataset.zip" -d /content/dataset
```

Step 11: Verify data.yaml

Code:

```
!cat /content/dataset/data.yaml
```

Step 12: Train YOLOv8 Model

Code:

```
from ultralytics import YOLO

model = YOLO("yolov8n.pt")

model.train(
    data="/content/dataset/data.yaml",
    epochs=30,
    imgsz=640,
    batch=16,
    device=0,
```

```
workers=2,  
cache=False  
)
```

Step 13: Locate Best Model

Code:

```
!find /content -name "best.pt"
```

Step 14: Download Best Model

Code:

```
from google.colab import files  
files.download("/content/runs/detect/train/weights/best.pt")
```

Step 15: Export to ONNX

Code:

```
from ultralytics import YOLO  
  
model = YOLO("/content/runs/detect/train/weights/best.pt")  
model.export(format="onnx", imgsz=640)
```

Step 16: Locate ONNX File

Code:

```
!find /content -name "*.onnx"
```

Step 17: Download ONNX File

Code:

```
from google.colab import files  
files.download("/content/runs/detect/train/weights/best.onnx")
```

Step 18: Prepare for RDKit X5

Workflow:

```
Images  
→ Roboflow Dataset  
→ YOLOv8 Training  
→ best.pt  
→ best.onnx  
→ Horizon Conversion  
→ RDKit X5 Deployment
```